Spike: Task 17

Title: Sprites and Graphics

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Goals / deliverables:

The goal of this spike task is to demonstrate how I load 2d images from file and display them on the screen using SDL2.

Besides this report, what else was created?

- Code files in see COS30031-2023-102608927\17 - Spike - Sprites and Graphics\Sprite and Graphics Task\Sprite and Graphics Task

Technologies, Tools, and Resources used:

List of information needed by someone trying to reproduce this work.

- Visual Studio 2022 Community Edition
- GitHub
- SDL2 Library

Tasks undertaken:

- Project setup for SDL2 library dependencies
- Create basic bmp images.

Initialization and Image Loading:

To begin the implementation, SDL's video subsystem was initialized with SDL_Init(SDL_INIT_VIDEO). This is a fundamental step as it prepares SDL for any graphics-related functionality.

```
if (SDL_Init(SDL_INIT_VIDEO) != 0) {
        std::cerr << "Error initializing SDL: " << SDL_GetError() << std::endl;
        return 1;
    }
```

Following this, I created a window using SDL_CreateWindow, which gives us a canvas on which graphics, such as images, can be rendered. An associated renderer was initialized using SDL_CreateRenderer. This renderer is pivotal as it facilitates the rendering of images onto the previously created window.

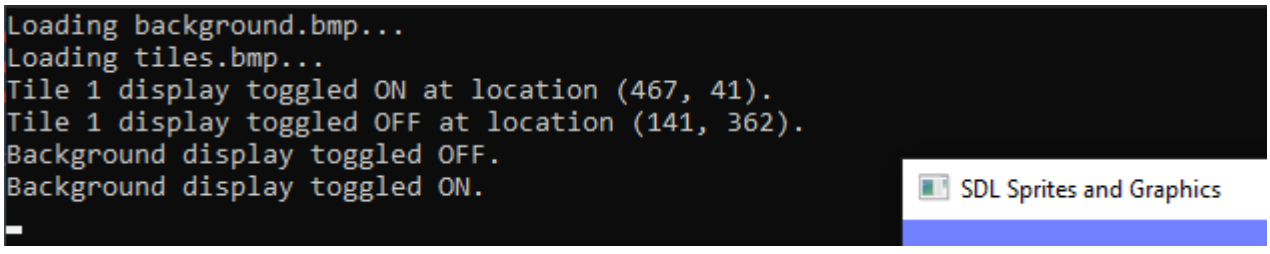
Displaying and Manipulating Images:

Once images are loaded, the next step is rendering them. For the background image, the renderer was used to ensure it occupied the entirety of the window. Additionally, I added a toggle using the “0”, which allows the user to make the background image visible or hidden on command.

```
1   SDL_Rect tileRects[3] = {
2           {0, 0, 50, 50},
3           {50, 0, 50, 50},
4           {100, 0, 50, 50},
5       };
```

Using the **SDL_Rect** structure. This structure is integral to defining specific sections of the loaded tileset image. Each sub-region (or tile) was extracted from the main image and rendered onto the screen at random positions. This rendering was made interactive by allowing users to toggle the visibility of each tile using the number keys. This is evident in the case structures where specific keys correspond to displaying specific tiles.

Debugging:



By using console logs, it was very instrumental when I was debugging. As the embedded messages like “loading image1.bmp” or “tile 3 display ON at location (10, 40)”, tracking the application's flow and diagnosing issues became considerably more manageable. Furthermore, it's important to highlight the concept of optimizing images to the current

screen's bit-depth. This optimization ensures that the rendering process is efficient, minimizing potential performance lags or visual artifacts.